

MODEL SOLUTION

Q1.

(A). Gain of the system,

$$K = \frac{10.2 - 7}{28 - 16} = \frac{0.2667 \text{ mm/kN}}{0.2667 \text{ mm/kN}} \quad (1M)$$

also, $m_p = \left(\frac{11.6 - 10.2}{10.2 - 7} \right) = 0.4375 \quad (1M)$

$$\therefore \xi = ? \Rightarrow \because e^{-\xi\pi/\sqrt{1-\xi^2}} = 0.4375$$

$$\Rightarrow -\xi\pi = \ln(0.4375) \times \sqrt{1-\xi^2}$$

$$\Rightarrow \xi^2 \pi^2 = (0.8267)^2 (1-\xi^2)$$

$$\Rightarrow \xi^2 (\pi^2 + 0.8267^2) = 0.8267^2 \Rightarrow \xi^2 = 0.0647 = \underline{\underline{0.254}}$$

$$\therefore \xi = 0.254 \quad (3M)$$

By graph, we have,

damped period of oscillation $(P) = 3.28 \text{ sec.}$

$$\text{so, } \because \omega_d = \omega_n \sqrt{1-\xi^2}$$

$$\Rightarrow \frac{2\pi}{P} = \omega_n \sqrt{1-\xi^2} \Rightarrow P \omega_n = \frac{2\pi}{\sqrt{1-\xi^2}}$$

$$\Rightarrow \omega_n = 2.03 \text{ rad/sec.} \quad (3M)$$

$$\begin{aligned} \therefore G(s) &= \frac{Y(s)}{X(s)} = \frac{0.2667 \times \frac{2.03^2}{(s^2 + 2 \times 2.03 \times 0.254s + 2.03^2)}}{\frac{1.099}{s^2 + 1.031s + 4.121}} \\ &= \frac{1.099}{s^2 + 1.031s + 4.121} \quad (2M) \end{aligned}$$

(B) - $\therefore G(s) = \frac{1.099}{s^2 + 1.031s + 4.121}$

& $G_{CL}(s) = \frac{1.099}{s^2 + 1.031s + 4.121 + 1.099 K_D s}$

$\therefore (1.031 + 1.099 K_D) = 2\xi'\omega_n$

$\therefore \xi' = 1/\sqrt{2}$

So, $1.031 + 1.099 K_D = \sqrt{2} \times \sqrt{4.121}$

$\Rightarrow K_D = \underline{\underline{1.674}}$

5 M

5 M

(C) - for open loop system,

$G(s) = \frac{1.099}{s^2 + 1.031s + 4.121}$

$\therefore \phi = -\tan^{-1}\left(\frac{1.031\omega}{4.121 - \omega^2}\right) \Rightarrow \phi = -\tan^{-1}\left(\frac{1.031}{3.121}\right)$
 $= -18.28^\circ$

$\Rightarrow \text{time lag} = \frac{2\pi}{360^\circ} \times (18.28^\circ) = 0.32 \text{ sec}$

$\textcircled{2\frac{1}{2} M}$

for, closed loop configuration,

$G_{CL}(s) = \frac{1.099}{s^2 + 2.878s + 4.121}$

$\therefore \phi = -\tan^{-1}\left(\frac{2.878\omega}{4.121 - \omega^2}\right) = 42.6^\circ$

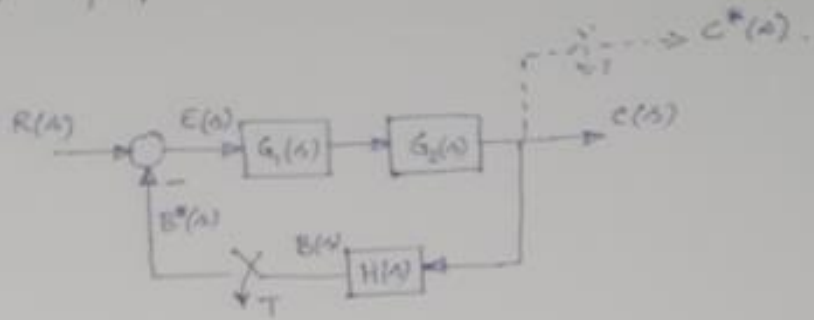
$\therefore \text{time lag} = \frac{2\pi}{360^\circ} \times 42.6^\circ = 0.743 \text{ sec}$

$\textcircled{2\frac{1}{2} M}$

Q2.

Q2. let us apply superposition theorem.

i) $D=0$.



clearly, $E(s) = R(s) - B^*(s)$.

$$\text{Now, } B(s) = E(s) \cdot G_1(s) G_2(s) H(s).$$

$$= \{R(s) - B^*(s)\} G_1(s) G_2(s) H(s)$$

$$= R(s) G_1(s) G_2(s) H(s) - B^*(s) G_1(s) G_2(s) H(s)$$

$$\therefore B^*(s) = R G_1 G_2 H^*(s) - B^*(s) \cdot G_1 G_2 H^*(s)$$

$$\Rightarrow B^*(s) = \frac{R G_1 G_2 H^*(s)}{1 + G_1 G_2 H^*(s)}$$

$$\text{Now, } C(s) = G_1(s) G_2(s) \cdot E(s)$$

$$= G_1(s) G_2(s) \{R(s) - B^*(s)\}.$$

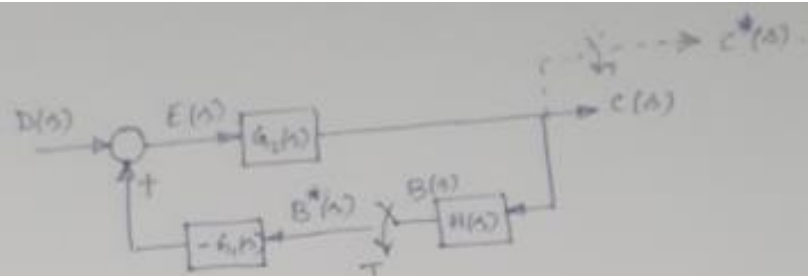
$$\Rightarrow C^*(s) = G_1 G_2 R^*(s) - G_1 G_2^*(s) B^*(s).$$

$$= G_1 G_2 R^*(s) - \frac{G_1 G_2^*(s) \cdot R G_1 G_2 H^*(s)}{1 + G_1 G_2 H^*(s)}.$$

$$\therefore C(z) = G_1 G_2 R(z) - \frac{G_1 G_2(z) \cdot G_1 G_2 H R(z)}{1 + G_1 G_2 H(z)}.$$

9

ii) $R=0$.



$$\begin{aligned} C(s) &= G_2(s) \cdot E(s) \\ &= G_2(s) \{ D(s) - G_1(s) B^*(s) \} \\ &= G_2(s) D(s) - G_1(s) G_2(s) B^*(s) \end{aligned}$$

$$\begin{aligned} \text{Now, } B(s) &= E(s) G_2(s) H(s) \\ &= \{ D(s) - G_1(s) B^*(s) \} G_2(s) H(s) \\ &= G_2(s) H(s) D(s) - G_1(s) G_2(s) H(s) B^*(s) \end{aligned}$$

$$\Rightarrow B^*(s) = G_2 H D^*(s) - G_1 G_2 H^*(s) B^*(s)$$

$$\Rightarrow B^*(s) = \frac{G_2 H D^*(s)}{1 + G_1 G_2 H^*(s)}$$

$$\text{Now, } C(s) = G_2(s) D(s) - G_1(s) G_2(s) B^*(s)$$

$$\Rightarrow C^*(s) = G_2 D^*(s) - G_1 G_2^*(s) \cdot \frac{G_2 H D^*(s)}{1 + G_1 G_2 H^*(s)}$$

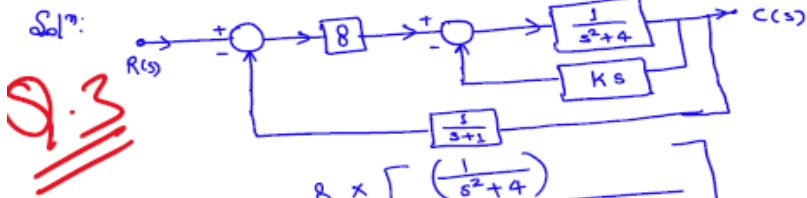
$$\Rightarrow C(z) = G_2 D(z) - \frac{G_1 G_2(z) \cdot G_2 H D(z)}{1 + G_1 G_2 H(z)} \quad (9)$$

When both R & D are present,

$$C(z) = G_1 G_2 R(z) + G_2 D(z) - \frac{G_1 G_2(z)}{1 + G_1 G_2 H(z)} [G_1 G_2 H R(z) + G_2 H D(z)]$$

Q3.

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$$\frac{C(s)}{R(s)} = \frac{8 \times \left[\frac{\left(\frac{1}{s^2+4} \right)}{1 + \left(\frac{1}{s^2+4} \right) (Ks)} \right]}{1 + \frac{8 \times \left(\frac{1}{s^2+4} \right)}{1 + \left(\frac{1}{s^2+4} \right) (Ks)} \times \left(\frac{1}{s+1} \right)}$$

$$\frac{C(s)}{R(s)} = \frac{\left(\frac{8}{s^2+4+Ks} \right)}{1 + \left(\frac{8}{s^2+4+Ks} \right) \left(\frac{1}{s+1} \right)}$$

$$\frac{C(s)}{R(s)} = \frac{8(s+1)}{(s+1)(s^2+4+Ks) + 8}$$

Characteristic eqⁿ: $(s+1)(s^2+4+Ks) + 8 = 0$

$$(s+1)(s^2+4) + (s+1)Ks + 8 = 0$$

$$1 + \frac{K(s+1)s}{(s+1)(s^2+4) + 8} = 0$$

$$1 + G(s)H(s) = 0$$

∴ open-loop T.F.

$$G(s)H(s) = \frac{K(s+1)s}{(s+1)(s^2+4) + 8}$$

$$G(s)H(s) = \frac{K(s+1)s}{(s+1)(s^2-s+4)}$$

3 M

5 M

① $n_p = 3$; $n_z = 2$

② $P = -2$; $\frac{1 \pm j\sqrt{23}}{2} \approx 0.5 \pm j2.4$

$z = 0, -1$

③ Centroid : $-\sigma_A = \frac{\sum P - \sum Z}{n_p - n_z}$

$$-\sigma_A = \frac{-2 + 0.5 + 0.5 - (0 - 1)}{3 - 2}$$

$$-\sigma_A = 0$$

Asymptotes emerges from '0'

④ Angle of asymptotes: ϕ_A

$$\phi_A = \left(\frac{2q+1}{n_p - n_z} \right) 180^\circ \quad q = 0, 1$$

$$\phi_A = \frac{180}{3-2} = 180^\circ, \quad \frac{3}{3-2} \times 180^\circ = 540^\circ$$

3

(5) Intersection on $j\omega$ -axis

char eq: $1 + \frac{k(s+2)s}{(s+2)(s^2-s+6)} = 0$

$$s^3 - s^2 + 6s + 2s^2 - 2s + 12 + ks^2 + ks = 0$$

$$s^3 + (1+k)s^2 + (4+k)s + 12 = 0$$

s^3	1	$4+k$	0
s^2	$1+k$	12	0
s^1	$\frac{(1+k)(4+k)-12}{1+k}$	0	
s^0	12		

$$\frac{(1+k)(4+k)-12}{1+k} > 0$$

$$k^2 + 4k + k + 4 - 12 > 0$$

$$k^2 + 5k - 8 > 0$$

$$k = \frac{-5 \pm \sqrt{25+32}}{2} = \frac{-5 \pm 7.55}{2}$$

$$k = 1.275, -6.275$$

for stable system $k^2 + 5k - 8 > 0$
i.e. $k < 1.275$

For $k = 1.275$ coefficients of $s^1 \approx 0$

Auxiliary eq.

$$(1+k)s^2 + 12 = 0$$

$$(1+1.275)s^2 + 12 = 0$$

$$j\omega = s = \sqrt{\frac{-12}{2.275}} = \pm j 2.2966$$

(6) Break-away point

$$f(s) = s^3 + (1+k)s^2 + (4+k)s + 12 = 0$$

$$\frac{dk}{ds} = \frac{d}{ds} \left(\frac{s^3 - s^2 - 4s - 12}{s^2 + s} \right)$$

$$s^4 + 2s^3 - 3s^2 - 24s - 12 = 0$$

clearly, Break-away or break in point lies

b/w 0 & -1 hence for trial let $s = -0.5$

$$f(-0.5) = -0.0289, \text{ Hence } f(s) = 0 \text{ is}$$

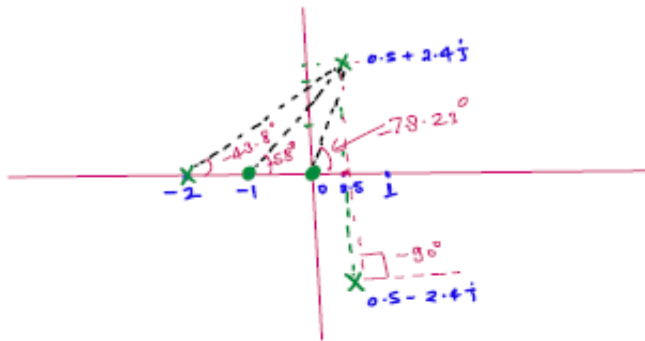
close to $s = -0.5$ choose $s = -0.55$

$$f(-0.55) \text{ or } f(-0.6) \dots$$

Hence $\rightarrow \boxed{s = -0.5474} \checkmark$

⑦ Angle of departure: θ_d

②



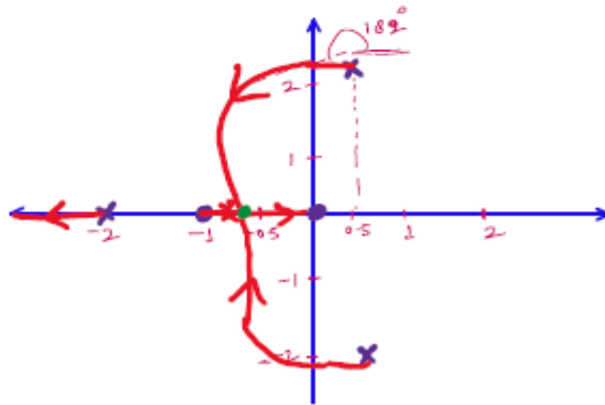
$$\theta_d = 180^\circ + [\theta_p - \theta_z]$$

$$\theta_d = 180^\circ + [-90^\circ - 43.8^\circ - (-58^\circ - 78.23^\circ)]$$

$$\theta_d = 182.43^\circ$$

Now draw RL with this calculation

④



For underdamped response:

find range of 'k' b/w break-away & jw-intersection points.

$$\left| \frac{k(s+2)s}{(s+2)(s^2-s+6)} \right| = 1$$

$$k = \left| \frac{(s+2)(s^2-s+6)}{(s+1)(s)} \right|$$

$$k = \left| \frac{(s+2)(s^2-s+6)}{(s+1)(s)} \right|$$

$$\left. \begin{array}{l} \text{at } s = -0.5474 \text{ \& } \\ s = \pm 2.2966 \end{array} \right\}$$

$$1.29 < k < 40.1 \quad \in \text{ Roots are complex.}$$

4M

Q4.

Q.4

Q.2 Given:

$$G(s) H(s) = \frac{k(s-1)}{s(s+1)}$$

$$\begin{aligned} s &= j\omega \\ G(j\omega) H(j\omega) &= \frac{k(j\omega-1)}{j\omega(j\omega+1)} \\ &= \frac{k(j\omega-1)(1-j\omega)}{j\omega(1+\omega^2)} \\ &= \frac{k(1-j\omega)(1-j\omega)j}{\omega(1+\omega^2)} \\ &= \frac{k(1-\omega^2-j2\omega)j}{\omega(1+\omega^2)} \\ &= \frac{2k}{(1+\omega^2)} + j \frac{k(1-\omega^2)}{\omega(1+\omega^2)} \end{aligned}$$

4 M

$$\begin{aligned} |G(j\omega) H(j\omega)| &= \left(\frac{k}{1+\omega^2} \right) \left| \left[2 + j \frac{(1-\omega^2)}{\omega} \right] \right| \\ &= \left(\frac{k}{1+\omega^2} \right) \sqrt{4 + \frac{(1-\omega^2)^2}{\omega^2}} \\ &= \left(\frac{k}{\omega(1+\omega^2)} \right) \sqrt{4\omega^2 + 1 + \omega^4 - 2\omega^2} \\ &= \frac{k}{\omega(1+\omega^2)} \sqrt{\omega^4 + 2\omega^2 + 1} \\ &= \frac{k}{\omega} \times \frac{1+\omega^2}{1+\omega^2} \\ &= \frac{k}{\omega} \end{aligned}$$

2 M

$$\begin{aligned} \angle G(j\omega) H(j\omega) &= -90^\circ - \tan^{-1} \omega + (180^\circ - \tan^{-1} \omega) \\ &= 90^\circ - 2 \tan^{-1} \omega \end{aligned}$$

2 M

$$(i) \lim_{\omega \rightarrow 0} G(j\omega) H(j\omega) = 2k + j\infty$$

$$(ii) \lim_{\omega \rightarrow \infty} G(j\omega) H(j\omega) = 0 \angle -90^\circ = 0 - j0$$

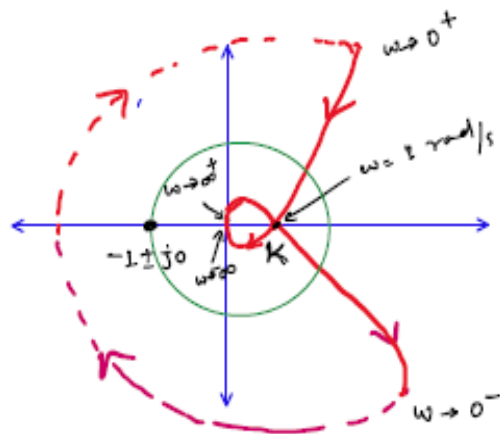
(iii) Intersection with Re & Im-axis

$$a) \text{Re} = 0 = \frac{2k}{1+\omega^2} = 0$$

No intersection with Im-axis

$$b) \text{Im} = 0 = \frac{k(1-\omega^2)}{\omega(1+\omega^2)} = 0$$

Intersect at Re-axis $\Rightarrow \omega = 1 \text{ rad/s}$



Real axis is intersected at $= \frac{k}{1} = k$
 ORIGIN POLE Test: $\omega = 1 \text{ rad/s}$

$$G(s)H(s) \Big|_{s=j\omega} = \lim_{\omega \rightarrow 0} \frac{-k}{\omega e^{j\theta}} = \lim_{\omega \rightarrow 0} \frac{-k}{\omega e^{j\theta}} = \infty \angle -180^\circ - \theta$$

From open-loop T.F. $P = 0$
 $Z = N + P$ from plot $N = 1$
 $Z = 1$

For any value of 'k' C.L. is unstable

6M

6M

2

Q5.

$$1) G(s) = \frac{1}{s+2}$$

$$OLTF = \frac{K}{s(s+2)(s+4)} ; G(j\omega)H(j\omega) = \frac{K}{j\omega(j\omega+2)(j\omega+4)}$$

ii) PM is given as 45°

$$\text{So } \angle GH|_{\omega_c} \neq 180^\circ = 45^\circ \Rightarrow \angle GH|_{\omega_c} = -135^\circ$$

$$\angle GH = -90^\circ - \tan^{-1} \omega/2 - \tan^{-1} \omega/4$$

$$\tan^{-1} \frac{\omega}{2} + \tan^{-1} \frac{\omega}{4} = 45^\circ$$

$$\frac{\frac{\omega}{2} + \frac{\omega}{4}}{1 - \frac{\omega^2}{8}} = 1 \Rightarrow 8 - \omega^2 = 6\omega$$

$$\Rightarrow \omega^2 + 6\omega - 8 = 0$$

$$\omega = \frac{-6 \pm \sqrt{36 + 32}}{2} = \frac{-6 \pm \sqrt{68}}{2} \Rightarrow \omega = 1.123 \text{ rad/s}$$

$$\frac{K}{\omega(\omega^2+4)(\omega^2+16)} = 1 \Rightarrow K = 10.7$$

$$\text{For this } K(10.7) \quad \omega_p \Rightarrow -\tan^{-1} \omega_p/2 - \tan^{-1} \omega_p/4 - 90^\circ = -180^\circ$$

$$\Rightarrow \frac{\omega_p/2 + \omega_p/4}{1 - \omega_p^2/8} = \infty \Rightarrow 8 - \omega_p^2 = 0 \Rightarrow \omega_p = 2.828 \text{ rad/s}$$

$$|GH|_{\omega_p} \Rightarrow \frac{10.7}{\omega_p(\omega_p^2+4)(\omega_p^2+16)} = 0.273$$

$$\Rightarrow 20 \log 0.273 = -11.276$$

$$(c) \text{ ess} = \frac{1}{K_v} \Rightarrow \frac{8}{K} = \frac{8}{10.7} = 0.7476$$